

MVP5 at the Frontier Occurrence Lift: A Source-Locked Route Decision after the Four-Map and Actual-Core Audits

Liang Wang
School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology, Wuhan 430074, P.R. China
wangliang.f@gmail.com

July 2026

Abstract

MVP4 located the first unsupported carrier after a complete cut. TPC-143–151 now determine what that statement means path by path. The four downstream maps share one first prerequisite: a conservative row-separated occurrence lift from every eligible-tail and frontier cut path to fully typed downstream occurrences. Abstract quotient, kernel and commuting-square tests are proved, but the actual lift and all actual downstream maps remain unavailable.

The same batch contains a genuine arithmetic advance. An exact quotient identity places a fixed-two Möbius periodic core in a small-polylogarithmic almost-scale correlation theorem. We retain this as $L1_{\text{actual-core}}$ evidence. It is not positive L2: the full physical scope and a fixed X -power saving are absent.

We freeze the combined state in an MVP5 classifier. Validation precedes seven ordered route verdicts. Architecture infeasibility requires a theorem-backed complete route universe; rerouting requires a fresh registry and typed crosswalk; a route stop requires exact complete-cell scope; and the arithmetic frontier requires every structural gate, an arithmetically independent H9 registry and a strict physical-loss endpoint. The current source-locked verdict is

$$\boxed{\text{Verdict}_{143:151} = \text{NOT-TESTABLE}},$$

with minimal missing set $\{\text{H1.frontier_occurrence_lift}\}$. This is forward structural and actual-core progress, but no hard-packet, prime-pair or twin-prime theorem.

Keywords: fixed shift; occurrence lift; route decision; actual Möbius core; arithmetic frontier; endpoint certificate.

Claim boundary. NOT-TESTABLE means that an active typed artifact is unavailable; it does not mean the selected mathematical route is false. The stopped cells below are narrower architectures, not a complete impossibility theorem. A real almost-scale Möbius core theorem is not the full physical H3 theorem. No positive fixed- X -power L2 result, strict endpoint, $B_{h0,\delta} = o(X)$, prime-pair lower bound, or twin-prime theorem is claimed.

1 From MVP4 to MVP5

MVP4 moved the first-failure pointer from a missing native archive to a frontier-totalization node after an exact three-way cut [1, 2]. TPC-143–146 refine that node into an executable four-map contract [3, 4]. Its first object is

$$L_X : \mathbb{C}^{\mathcal{C}_X^{\text{ns}}} \longrightarrow \mathbb{C}^{\mathcal{O}_X}, \quad \mathbf{1}^T L_X = \mathbf{1}^T,$$

where

$$\mathcal{C}_X^{\text{ns}} = \mathcal{C}_X^{\text{ETO}} \sqcup \mathcal{C}_X^{\text{FUM}}.$$

The actual frozen sample has no ETO row and 2,988 FUM rows, but the required domain remains both classes.

The structural batch proves:

- (i) $P_{h_0}^{\text{cut}} = I$, with an explicit prohibition against calling it the downstream selector;
- (ii) field-descent, Q_D/Q_Z kernel and G/P_{h_0} occurrence-level commutation criteria;
- (iii) a zero-defect contract for the completed four-map route; and
- (iv) scoped stops for current-schema-only and aggregate-cancellation implementations.

Here “current-schema-only” means the maximal formal completion class constrained by the archived fields and identities. It is not an impossibility theorem on the unknown actual-support carrier. It does not construct L_X . The independent scalar route is not closed either: it requires both a complete original-frontier $o(X)$ theorem and a theorem-backed disposition of every ETO path. Thus the empty ETO class in the finite fixture and a frontier scalar estimate alone cannot certify all-nonsoft totalization.

TPC-149 independently proves an actual fixed-two Möbius periodic core almost-scale estimate, and TPC-150 records the missing physical return and an atomic-prefix firewall [5, 6]. TPC-151 integrates the two branches without a promotion edge [7].

2 A valid source-locked snapshot

Definition 2.1 (Valid MVP5 snapshot). A snapshot is valid when:

- (a) its TPC-151 manifest and schema have the recorded canonical `CANONICAL_UTF8_LF_V2` hashes;
- (b) hashes are explicitly integrity-only;
- (c) every import matches status, level, role, direction, scope, carrier, normalization and coverage;
- (d) the proof DAG is acyclic and its displayed order is canonical;
- (e) the selected active set is derived from the selected-root ancestor closure;
- (f) the route and stop tables are exact typed covers of the *declared* tested cells;
- (g) H9 has no direct or indirect arithmetic-role or H2–H5 ancestor; and
- (h) all rational endpoint records and claim-boundary Booleans pass their direction checks.

Failure of one of these conditions returns `INVALID` before any mathematical route verdict. In particular, source drift makes the certificate stale; it does not refute a theorem.

3 The minimal missing antichain

Let $\mathcal{A}$ be the active ancestor closure of the selected root, including the root. Define

$$\mathcal{M} = \{v \in \mathcal{A} : \text{status}(v) = \text{NOT-TESTABLE or } v \text{ fails scope matching}\}.$$

The partial-order frontier of missing nodes is

$$\min(\mathcal{M}) = \{v \in \mathcal{M} : \text{no ancestor of } v \text{ belongs to } \mathcal{M}\}.$$

Only for deterministic reporting, the first representative is chosen by canonical topological rank and node identifier.

Proposition 3.1 (Frozen MVP5 missing frontier). *For the selected occurrence-lift/quotient-Möbius route,*

$$\boxed{\min(\mathcal{M}) = \{\text{H1.frontier_occurrence_lift}\}}.$$

Proof. The cut selector and abstract interface nodes are proved. The occurrence lift is NOT-TESTABLE. Every actual Q_D, Q_Z, G , downstream P_{h_0} , H1 carrier, H5–H9 and physical-return missing node is a descendant of it in the selected DAG. It is therefore the unique minimal missing node. $\square$

This algorithm prevents a later missing map from hiding its missing parent and avoids treating incomparable prerequisites as if they had a canonical mathematical order.

4 Eight outcomes and their precedence

The outer classifier has one validation outcome and seven valid-snapshot verdicts.

Definition 4.1 (Ordered valid-snapshot verdicts). After validation, test:

- (i) GO: all active required gates are proved in compatible scopes and the full-synthesis endpoint has strict positive slack.
- (ii) ARCHITECTURE-INFEASIBLE: an independent theorem proves the declared route universe complete and exact complete-cell negative certificates stop every route.
- (iii) REROUTE: the selected route is stopped and a distinct open typed alternative, fresh registry and theorem-backed crosswalk are selected.
- (iv) STOP-ROUTE: the selected route has a complete exact-cell stop certificate and no typed alternative is selected.
- (v) NOT-TESTABLE: an active required artifact is unavailable or scope-incompatible.
- (vi) ARITHMETIC-FRONTIER: every structural gate is proved, H9 is arithmetically independent, the physical-loss endpoint passes strictly, and every unresolved active node is an open scope-matched positive-L2 arithmetic target.
- (vii) OPEN: every remaining valid case.

Theorem 4.2 (Total deterministic classifier). *Validation followed by this precedence returns exactly one of*

INVALID, GO, ARCHITECTURE-INFEASIBLE, REROUTE,
STOP-ROUTE, NOT-TESTABLE, ARITHMETIC-FRONTIER, OPEN.

Proof. Validation either fails or provides a well-typed finite snapshot. On a valid snapshot the ordered list ends in a catch-all, so its first true predicate is unique. The precedence deliberately separates a complete architectural stop, a selected-cell stop, a missing object and the arithmetic frontier. $\square$

5 Strengthened stop and reroute rules

A list of tested routes is not automatically the universe of every possible argument. Therefore

all declared cells stopped $\not\Rightarrow$ ARCHITECTURE-INFEASIBLE

unless an independent source proves the list complete in the actual physical scope.

Likewise a negative result is stored with its exact scope, carrier, normalization, registry and coverage. The schema-only quotient obstruction does not stop an occurrence-augmented quotient route; an aggregate-commutator counterexample does not stop a row-separated lift; and an atomic-prefix obstruction does not stop every possible all-prefix theorem.

Finally, a reroute is not a change of a string identifier. The alternative must be open, use a distinct registry and possess a theorem-backed crosswalk. Otherwise the snapshot is invalid rather than REROUTE.

6 Three evidence labels

The batch requires three progress labels:

L1 _{structural} :	physical interfaces, typed covers and completion contracts,
L1 _{actual-core} :	a genuine arithmetic theorem in a strict core scope,
L2 _{actual-positive} :	a fixed- X -power saving on the literal physical carrier.

The TPC-149 theorem belongs to the middle line. It evaluates actual Möbius factors after an exact quotient lift and is stronger than a synthetic model. It remains below the third line because its scope is almost-scale and periodic-core, it lacks the complete physical return, and its fixed- X -power exponent is zero.

Proposition 6.1 (No pseudo-frontier promotion). *A conditional return formula, an $L1_{\text{actual-core}}$ theorem, a frozen-family estimate or a logarithmic saving cannot be the only unresolved evidence admitted by the ARITHMETIC-FRONTIER predicate.*

Proof. The predicate requires unresolved records to have status OPEN, evidence type $L2_{\text{target-positive}}$, exact actual scope and nonstructural role. Each listed substitute fails at least one required field. $\square$

7 The split 1/400 endpoint

MVP5 separates:

$$\begin{aligned} \text{arithmetic target} & : \sigma_{\text{raw}} \geq 1/400, \\ \text{physical-loss target} & : \Lambda_{\text{phys}} < 1/400, \\ \text{full synthesis} & : \sigma_{\text{raw}} - \Lambda_{\text{phys}} > 0. \end{aligned}$$

The ARITHMETIC-FRONTIER verdict requires the second line but deliberately leaves the first as an open arithmetic target. The GO verdict requires all three lines in compatible scope.

For the physical ledger, an upper certificate $U < 1/400$ is a strict pass. A complete lower or exact dual certificate $L \geq 1/400$, including equality, stops the corresponding route cell. A conservative upper certificate $U \geq 1/400$ without a lower certificate is neither pass nor stop. Unknown tokens are not zero, energy exponents are halved once, joint replacements cover disjoint primitive occurrences, selector cost is charged once and determinant reserve is not physical slack.

The current H9 registry is incomplete, so the physical and full ledgers are both incomplete. The log-power actual-core saving is recorded separately at fixed- X -power exponent zero.

8 H9 arithmetic independence

The H9 occurrence registry is a structural/physical-synthesis certificate. Its allowed ancestors are H1, H6, H7, H8 and independent physical cost sources. The audit first rejects any ancestor whose role is arithmetic shadow, arithmetic target or arithmetic negative, then independently rejects H2–H5 gate ancestors.

This two-key check matters because a node could otherwise be renamed outside an H3 namespace while still importing the target correlation. In particular, the actual-core theorem cannot be used to fill an unknown physical occurrence cost, and H5 determinant reserve cannot be registered as endpoint slack.

9 Frozen gate projection

The selected route is `occurrence_lift_quotient_mobius_return`. It has no stop certificate. Six narrower cells have typed negative records, but the route-universe completeness status is OPEN.

10 MVP5 decision

Theorem 10.1 (Source-locked MVP5 verdict). *For the canonical TPC-143–151 snapshot dated 27 July 2026,*

$$\boxed{\text{Verdict}_{143:151} = \text{NOT-TESTABLE}, \quad \min(\mathcal{M}) = \{\text{H1.frontier_occurrence_lift}\}.$$

The verdict is neither GO, ARITHMETIC-FRONTIER, ARCHITECTURE-INFEASIBLE, nor a route stop.

Table 1: MVP5 gate projection.

Gate	Status	Evidence target	First unresolved content
H1	NOT-TESTABLE	$L1_{\text{structural}}$	Conservative row-separated occurrence lift on ETO and FUM.
H2	OPEN	positive L2	Literal signed resonance replacement.
H3	OPEN	positive L2	Physical weight, generic phase, all-prefix selector, four-sign return, and a fixed- X -power saving.
H4	OPEN	positive L2	Complete original-scale high/ultra/boundary return.
H5	NOT-TESTABLE	positive L2	Actual determinant/zero pair on completed occurrence fibers.
H6	NOT-TESTABLE	$L1_{\text{structural}}$	Complete physical occurrence cover.
H7	NOT-TESTABLE	$L1_{\text{structural}}$	Actual downstream fixed- h_0 totality.
H8	NOT-TESTABLE	$L1_{\text{structural}}$	Exact hard-packet reconnection.
H9	NOT-TESTABLE	$L1_{\text{structural}}$	Complete independent registry and physical-loss upper certificate.

Proof. The canonical source and type checks pass, so the snapshot is not INVALID. The selected route is open, and no theorem proves the declared route list complete, excluding all stop verdicts. By [proposition 3.1](#), an active structural artifact is unavailable, so NOT-TESTABLE occurs before ARITHMETIC-FRONTIER and OPEN. Independently, H6–H9 and the physical endpoint are incomplete, while TPC-149 supplies $L1_{\text{actual-core}}$, not positive physical L2. $\square$

Corollary 10.2 (What moved). *The first-failure pointer moved from generic frontier totalization to one explicit sparse matrix with an exact domain, conservation law and lineage schema. The arithmetic evidence simultaneously moved from a more remote affine shadow to an actual Möbius periodic core. These are forward moves, but neither closes a fixed- h_0 physical positive-L2 gate.*

11 Regression suite and next forced step

The standard-library audit reaches all seven valid verdicts on synthetic typed snapshots and rejects:

1. a malformed route before verdict selection;
2. current-schema nonidentifiability promoted to actual-carrier impossibility;
3. architecture infeasibility without a route-universe theorem;
4. rerouting without a fresh registry and typed crosswalk;
5. conditional or actual-core pseudo-frontiers;
6. a stale or descendant-pre-empted first-missing pointer;
7. a frontier-totalization node whose artifact drops the ETO clause from the scalar route;
8. the positive-L2 target label reported as an achieved fixed-power result;
9. an actual-core log saving assigned a positive X -power;
10. direct or indirect arithmetic ancestors of H9; and
11. a physical or full endpoint pass with missing registry data.

On the selected occurrence-augmented map route, the next forced structural object is L_X , not another refinement of an aggregated scalar. A successful construction must cover both ETO and FUM, conserve every source column, record every target occurrence separately and preserve native, shift,

normalization and support provenance. It would then make the four-map defect vector testable. The distinct scalar-plus-ETO route remains open under its own complete two-clause contract.

On the arithmetic side, the next separate task is to return the actual periodic core through the literal physical weights, phases and four-sign packet with either a deterministic all-prefix theorem or a valid non-atomic selector. Only after the structural endpoint passes independently can the snapshot legitimately return ARITHMETIC-FRONTIER.

Established. The valid-snapshot calculus, strengthened route-stop rules, minimal-missing antichain, evidence classification, H9 firewall and split endpoint semantics are L0/L1 results.

Not established. No occurrence lift, complete four-map physical carrier, positive fixed- X -power L2 saving, endpoint pass, $B_{h_0, \delta} = o(X)$, prime-pair lower bound or twin-prime theorem is proved.

12 Conclusion

MVP5 keeps both gains visible without combining them into a claim that neither source proves. The structural route now has one concrete first matrix, and the arithmetic route has one real actual-core theorem. The bridge between them is still missing. The honest source-locked verdict is therefore NOT-TESTABLE, with a sharper and more actionable pointer than MVP4.

References

- [1] Liang Wang. Source-locked integration at the first unsupported carrier. TPC-141 manuscript, 2026.
- [2] Liang Wang. Mvp4 at the first unsupported carrier. TPC-142 manuscript, 2026.
- [3] Liang Wang. The frontier occurrence-lift contract. TPC-143 manuscript, 2026.
- [4] Liang Wang. An executable four-map completion certificate for frontier totalization. TPC-146 manuscript, 2026.
- [5] Liang Wang. A small-polylogarithmic determinant-two möbius corridor. TPC-149 manuscript, 2026.
- [6] Liang Wang. Returning the determinant-two möbius corridor. TPC-150 manuscript, 2026.
- [7] Liang Wang. Source-locked integration at the frontier occurrence lift. TPC-151 manuscript, 2026.